

FPGA-Native Neural-Augmented Audio Codec — BTP Project Plan

Status: candidate BTP topic, pivoted from `HALAC_Hybrid Adaptive Low-Latency Audio Codec.pdf` after its original novelty claim was found to overlap with existing work. **This document:** folds a two-track literature/feasibility research pass (run 2026-08-12) into a full project plan — the idea, the research gap it targets, the steps to actually build it, the research paper as a separate deliverable, and the resources required.

1. The Idea

Title: FPGA-Native Neural-Augmented Audio Codec

One-line pitch: Every published neural or hybrid audio codec reports latency measured on GPU/CPU. Nobody has demonstrated one actually running in real time on FPGA silicon with genuinely measured hardware latency. This project builds one — a small causal neural network layered on a classical codec core, deployed and measured on real FPGA hardware — with that measured latency number as the headline result.

Origin: this pivots away from the original HALAC idea (a from-scratch hybrid codec competing on bitrate/quality against LDAC/Opus/LC3), whose novelty claim didn't hold up against existing work — 3GPP EVS and MPEG-D USAC already do classifier-driven mode switching between speech- and music-optimized coding paths, and Qualcomm's aptX Adaptive/Lossless already do adaptive high-fidelity low-latency wireless streaming. The pivot keeps HALAC's classical DSP core (MDCT/psychoacoustic coding) but moves the research contribution from "smarter mode switching" to "getting a neural-augmented codec onto real constrained hardware with a number you can actually measure."

Core approach:

1. A small, fully causal neural network (LACE/NoLACE pattern — proven, already shipped in Opus 1.5, ~300K parameters) that **never touches raw audio directly** — it predicts filter coefficients that steer a classical LPC/MDCT codec's own prediction filters.
2. Quantize that network (INT8/INT16 fixed-point) and deploy it, alongside the classical codec core, onto real FPGA hardware.
3. Measure real, on-chip, end-to-end encode+decode latency — not a simulation, not a synthesis-tool estimate.

Why this connects to existing work already underway: this directly reuses skills being built in the separate SilentLink Honors project — FPGA toolchain experience, 8-bit-quantized MAC unit design, and real-time audio DSP (FDLMS, psychoacoustic modeling) — turning Honors and BTP into one continuous research arc instead of two disconnected efforts.

Team: Joel (primary), Adithya (secondary researcher), advisor Dr. Ananth A — same advisor as the Honors SilentLink project.

2. Research Gap

2.1 The gap being targeted

No neural or hybrid (classical DSP + neural network) audio/speech **codec** — meaning something with both an encoder and decoder that compresses arbitrary input audio to a bitstream and reconstructs it — has been demonstrated running in real time on FPGA with measured (not synthesized/estimated) hardware latency, anywhere, as of an Aug 2026 literature search covering ~75 targeted queries across two independent research passes.

2.2 What was actually found (prior art landscape)

Closest near-miss — read this one in full before finalizing scope: arXiv:2606.04221 (July 2026), "Feasibility of Time-Domain DNN-Based Speech Enhancement on Embedded FPGA for Hearing Aids." A causal 1.5M-parameter time-domain CNN (SuDoRM-RF++ 0.25x), deployed on an AMD Kria KV260 (Zynq UltraScale+ MPSoC), 16-bit fixed-point, **hand-written Vitis HLS** (no automated toolchain). Result: **9.7ms measured first-sample hardware latency**, PESQ 2.41 / STOI 0.93. This is speech *enhancement/denoising* — no bitstream, no compression — so it isn't directly competing prior art, but it's architecturally almost identical to what SilentLink is already attempting, making it simultaneously the best available methodology template and a signal that this adjacent space is being actively worked.

Category	What was found	Relevance
Neural audio, real measured hardware latency (not codecs)	TinyVocos (MCU, not FPGA); the KV260 hearing-aid paper above; Intel WaveNet-on-Stratix-10-NX whitepaper (unverifiable primary source, 403s)	Adjacent, none are compression codecs
Classical (non-neural) codecs on FPGA	Codec2 on Cyclone IV (IEEE CSCI 2021); G.729.1 on Altera Nios II (2009)	Confirms "codec on FPGA" as an engineering pattern is old — only the neural piece is missing
Learned codec on FPGA, real measured latency — for images	F-LIC (arXiv:2503.04832, March 2025), Xilinx ZCU102, 43 FPS, real measured PSNR/energy	Proves the exact pattern works for a sibling modality — the audio instantiation specifically is what's missing
Hybrid classifier + classical MDCT/LPC, on FPGA	Nothing found	Equally open, and likely the more tractable framing for a BTP scope
Simulation-only or unverifiable hardware claims	Spiking Vocos (arXiv:2509.13049) — no evidence of real neuromorphic hardware run	Excluded as not real prior art

2.3 Search coverage caveats — act on this before fully committing

No direct IEEE Xplore, Google Scholar, or ACM Digital Library access in this research pass; no patent database search (Qualcomm/MediaTek/Realtek/HiSilicon file audio-chip patents that don't surface in academic search); no non-English venues checked. **Recommendation:** before committing months of work, have Joel or Dr. Ananth run one institutional-access pass on "neural audio codec FPGA hardware accelerator" plus forward-citation-chasing from AudioDec/F-LIC/LPCNet on Google Scholar, and a quick

patent-landscape check on major audio-SoC vendors. This is exactly the kind of blind spot that already weakened the original HALAC novelty claim once (missed EVS/USAC/aptX Adaptive) — cheap insurance against a repeat.

3. Project Completion Steps (Implementation Roadmap)

Ordered by dependency, not locked to specific calendar dates.

Phase 1 — Classical Codec Core (Software)

- Implement the classical LPC/MDCT codec baseline in software (this reuses HALAC's MDCT/psychoacoustic groundwork and Honors-project DSP experience).
- Verify correctness: lossless reconstruction before quantization, then baseline quantized quality with no neural component at all — this is the "classical-only" comparison point for every later result.

Phase 2 — Neural Side-Network (Software, Floating-Point)

- Implement a LACE/NoLACE-pattern network: small, fully causal, predicts filter coefficients (not raw audio) over the Phase 1 codec core.
- Train on a labeled/unlabeled mix of speech and music audio (see §5 for dataset list).
- Validate in floating-point simulation: does the neural-augmented path measurably beat the classical-only baseline on objective quality metrics (PESQ/STOI/PEAQ)? This must be true before spending any effort on hardware deployment.

Phase 3 — Quantization

- Quantize the trained network to INT8/INT16 fixed-point (quantization-aware training via Brevitas, or manual fixed-point conversion).
- Re-validate quality in simulation post-quantization — confirm the coefficient-prediction design tolerates quantization noise as expected (this is the specific risk the LACE-style architecture was chosen to avoid; see §3 of the original research notes for why raw-waveform-synthesis architectures like LPCNet/FARGAN were ruled out here).

Phase 4 — Hardware Deployment

- Select board: PYNQ-Z2 (cheaper, best student documentation, sufficient for a ~72K-weight-class network) or Kria KV260 (matches the closest real precedent exactly, comfortable headroom for the ~300K-parameter LACE-sized network) — see §5 for cost/spec detail.
- Implement the classical codec core in RTL/HLS, reusing SilentLink's FPGA toolchain where possible.
- Implement the quantized neural side-network via **hand-written Vitis HLS** — published practice in this exact niche (the KV260 hearing-aid paper) did not use an automated toolchain (hls4ml/FINN don't handle this class of model well; see the original feasibility research notes for the specific tooling gaps), so budget engineering time accordingly rather than assuming a push-button flow.
- Integrate the encoder+decoder pipeline on-chip, streaming, matching real audio sample rates.

Phase 5 — Measurement & Evaluation

- Measure real, on-chip, end-to-end encode+decode latency (the headline number).
- Measure resource utilization (LUTs, DSP slices, BRAM/URAM) and power draw.

- Run objective quality metrics (PESQ/STOI, and PEAQ if feasible) on hardware-processed audio, compared against: (a) the Phase 1 classical-only baseline on the same hardware, and (b) CPU/GPU-only versions of the same neural-augmented pipeline, to make the "hardware-native" contribution concrete and measurable.

Phase 6 — Optimization

- Tune the latency/quality trade-off (e.g., weight-caching strategy — on-chip BRAM capacity for weight storage was found to be the binding constraint in the closest real precedent, not raw DSP compute).
- Iterate on anything that falls short of target latency or quality.

4. Research Paper Plan (Separate Deliverable)

The paper is a distinct deliverable from the implementation above — drafting starts once Phase 5 produces a real measured latency number, since that number is the paper's central claim.

Structure:

1. **Introduction** — the gap (§2.1): every published neural/hybrid codec latency figure is GPU/CPU-measured, not real-hardware-measured
2. **Related Work** — classical FPGA codecs (Codec2, G.729.1), learned-codec-on-FPGA for images (F-LIC), the closest near-miss (arXiv:2606.04221 — explicit differentiation: enhancement vs. compression), LACE/NoLACE as the architectural basis
3. **Method** — codec core design, neural side-network architecture and training, quantization approach, hardware implementation (board, HLS/RTL structure, pipeline integration)
4. **Evaluation** — measured hardware latency (headline figure), resource utilization table, power draw, objective quality metrics vs. classical-only and CPU/GPU baselines
5. **Discussion** — what made real-hardware deployment hard in practice (tooling gaps, quantization behavior actually observed vs. predicted), limitations
6. **Conclusion & Future Work** — multi-channel, other codec cores, closing the loop with the SilentLink Honors project

Key figures/tables the paper needs:

- Architecture/pipeline block diagram
- Latency comparison table: this work (real hardware) vs. CPU/GPU baselines vs. the closest FPGA precedent (arXiv:2606.04221)
- Resource utilization table (LUTs/DSP/BRAM/URAM, power)
- Quality metric table (PESQ/STOI) — neural-augmented vs. classical-only, same hardware

Before finalizing the novelty paragraph: complete the institutional-access literature check from §2.3, and read arXiv:2606.04221 in full.

Deliverables checklist:

- [] Classical codec core, software baseline, verified
- [] LACE-pattern neural side-network trained, floating-point quality validated
- [] Quantized (INT8/INT16) network, quality re-validated

- [] Full encoder+decoder pipeline running on FPGA
 - [] Measured hardware latency, resource utilization, power draw
 - [] Objective quality metrics vs. classical-only and CPU/GPU baselines
 - [] Institutional-access literature re-check (§2.3) completed
 - [] Paper draft (Intro → Conclusion)
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5. Resources Required

Hardware: PYNQ-Z2 (~\$150–250, Zynq-7020, ~53K LUTs / 220 DSP / ~4.9Mb BRAM) or Kria KV260 (~\$200–450, Zynq UltraScale+, ~230K LUTs / ~1.2–1.7K DSP / ~11Mb BRAM + 27Mb URAM — matches the closest real precedent exactly). **Check first** whether a board already acquired for the SilentLink Honors project can be shared/reused before purchasing a second one.

Software/tools: Vitis HLS / Vivado for FPGA implementation; PyTorch + Brevitas for training and quantization-aware training of the neural side-network; Opus reference source code as the LACE/NoLACE architectural and comparison baseline; PEAQ/PESQ/STOI implementations for objective evaluation.

Datasets (training and evaluation audio, reused from the original HALAC resource list): EBU SQAM (Sound Quality Assessment Material), speech corpora (LibriSpeech, VCTK), diverse music samples. Total volume ~50GB expected.

Compute: standard development workstation (16GB+ RAM); GPU helpful but not strictly required for training given the neural side-network is small (~300K parameters, unlike a full end-to-end neural codec).

Team: Joel (primary), Adithya (secondary researcher), advisor Dr. Ananth A.

Evaluation logistics: objective metrics (PESQ/STOI/PEAQ) are sufficient for the core claims in this scope. A lightweight subjective listening comparison is optional and can be added later if time permits — the full 20+-subject MUSHRA infrastructure from the original HALAC plan is not required to support this project's headline claim (measured hardware latency), which lowers logistics burden relative to the original HALAC scope.